

Haplotyping as Perfect Phylogeny

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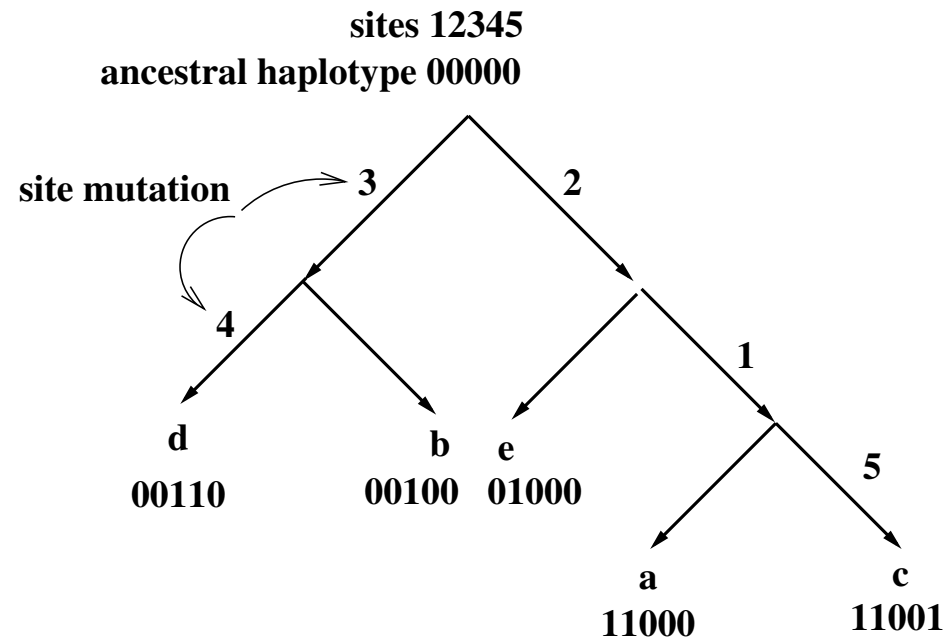
Date: Dec. 3, 2004

References

- Dan Gusfield., *Efficient Algorithms for Inferring Evolutionary trees*. 1991
- Dan Gusfield., *Haplotyping as Perfect Phylogeny: Conceptual Framework and Efficient Solutions*. 2002
- Vineet Bafina, Dan Gusfield, Giuseppe Lancia and Shibu Yooseph. *Haplotyping as Perfect Phylogeny: A Direct Approach*. 2002

Perfect Phylogeny

● The Perfect Phylogeny Model



● Matrix M has a phylogeny tree.

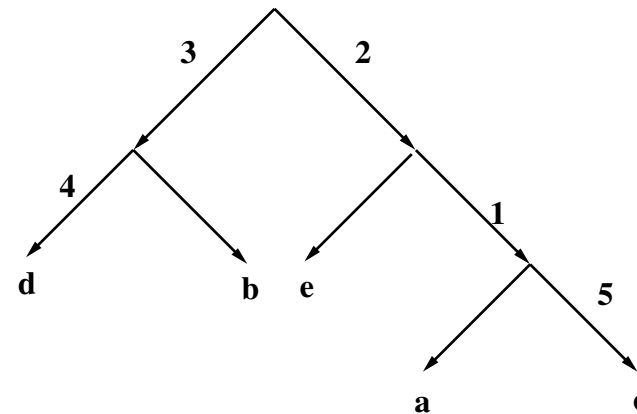
	1	2	3	4	5
a	1	1	0	0	0
b	0	0	1	0	0
c	1	1	0	0	1
d	0	0	1	1	0
e	0	1	0	0	0

Perfect Phylogeny (Contd.)

- Phylogeny Problem:** Given the n by m 0-1 matrix M , determine whether there is a phylogenetic tree for M , and if so, build one.

	1	2	3	4	5
a	1	1	0	0	0
b	0	0	1	0	0
c	1	1	0	0	1
d	0	0	1	1	0
e	0	1	0	0	0

→



- The Classical Theorem of Perfect Phylogeny:** The Matrix M has a perfect phylogeny. $\Leftrightarrow$ For each pair of columns, there are no three rows with values 0,1; 1,0; and 1,1 in those two columns.

1	1
1	0
0	1

- An $O(mn)$ Algorithm

 $\rightarrow 1$

→ 3

	2	1	3	5	4
$L(j)$	0	2	0	1	3

 $\rightarrow_4$ true

Testing a Phylogenetic Tree (Contd.)

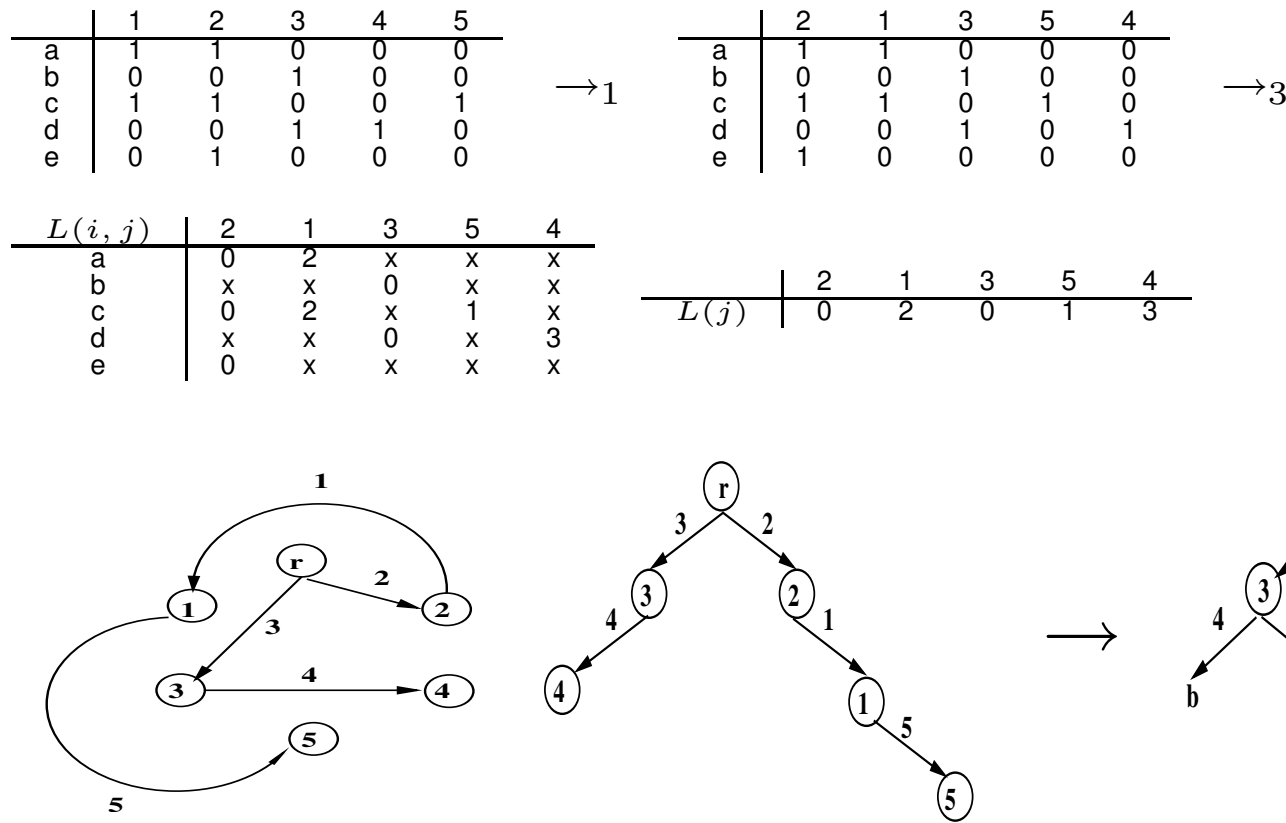
Algorithm

1. Considering each *columns* of M as a binary number, sorting these number into decreasing order.
2. Delete duplicate columns.
3. Compute $L(i, j)$ and $L(j)$.
4. Check whether $L(*, j) = L(j)$. If so then the answer is YES, otherwise NO.

- ## The time complexity
- $O(nm)$ by using radix sort in step 1.

Constructing a Phylogeny

- The $O(nm)$ time algorithm constructs a phylogeny.



Genotypes and Haplotypes

- At each site, each chromosome has one of two alleles (states) denoted by 0 and 1 (Motivated by SNPs).

1	1	0	0	0	0	1
1	0	1	1	0	0	1

Two Haplotypes per individual

Merge the haplotypes

1 2 2 2 0 0 1

Genotype for the individual

- Biological Problem:
Haplotype is more valuable than genotype data, but haplotype data is hard to collect. Genotype is easy to collect.

The HI Problem

- The Haplotype Inference (HI) problem:
Given a set S of n genotype vectors, find n pairs of haplotype vectors that could have generated the genotype vectors.
- Let M denote an n by m genotype matrix with $M[i, j] \in \{0, 1, 2\}$.
- If each i has h_i polymorphic sites, then there are $\prod_{i=1}^n 2^{h_i-1}$ solutions to the HI problem.
- Example:

	1	2	3
A	2	2	0
B	1	2	0
C	2	0	2

→

	1	2	3
a	1	1	0
a'	0	0	0
b	1	1	0
b'	1	0	0
c	1	0	0
c'	0	0	1

,

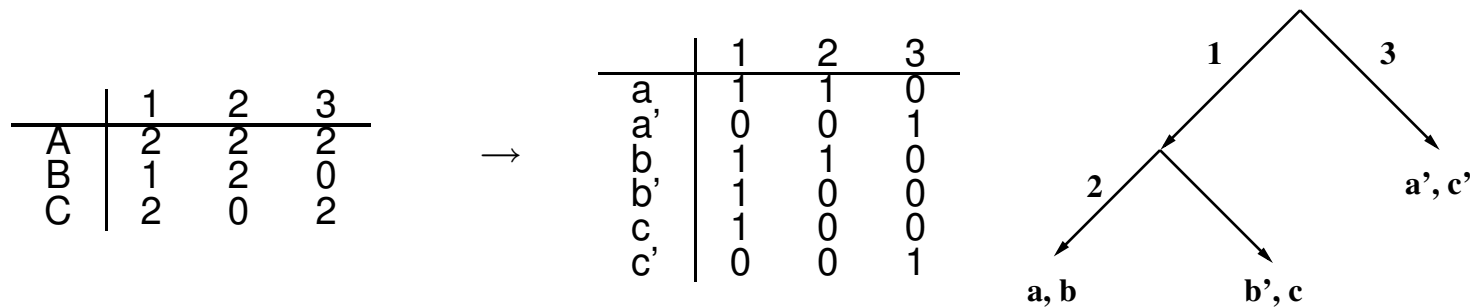
	1	2	3
a	1	0	0
a'	0	1	0
b	1	1	0
b'	1	0	0
c	1	0	0
c'	0	0	1

,

...

The HPP Problem

- The Haplotype Phylogeny Problem (PPH):
Given a set S of n genotype vectors, find an explaining set of haplotypes that fits a perfect phylogeny.



- Solving the PPH in $O(nm\alpha(nm))$ time. (Gusfield, RECOMB, Apr. 2002)
- A direct approach solves the PPH in $O(nm^2)$ time. (V. Bafna, D. Gusfield, G. Lancia and S. Yooseph, JVC, Feb. 2003)

Considering Two Columns

- Consider two columns j_{c1}, j_{c2} in a genotype matrix M .

	j_{c1}	j_{c2}
i_{r1}	2	2
i_{r2}	x	y

- j_{c1}, j_{c2} in a genotype are *compansions*, if there exists a row i such that $M[i, j_{c1}] = M[i, j_{c2}] = 2$.

$$\begin{matrix} 2 & 2 \end{matrix} \rightarrow \begin{matrix} 1 \\ 0 \end{matrix} \begin{matrix} 1 \\ 0 \end{matrix}$$

$$\begin{matrix} 2 & 2 \end{matrix} \rightarrow \begin{matrix} 1 \\ 0 \end{matrix} \begin{matrix} 0 \\ 1 \end{matrix}$$

Forced Compansions

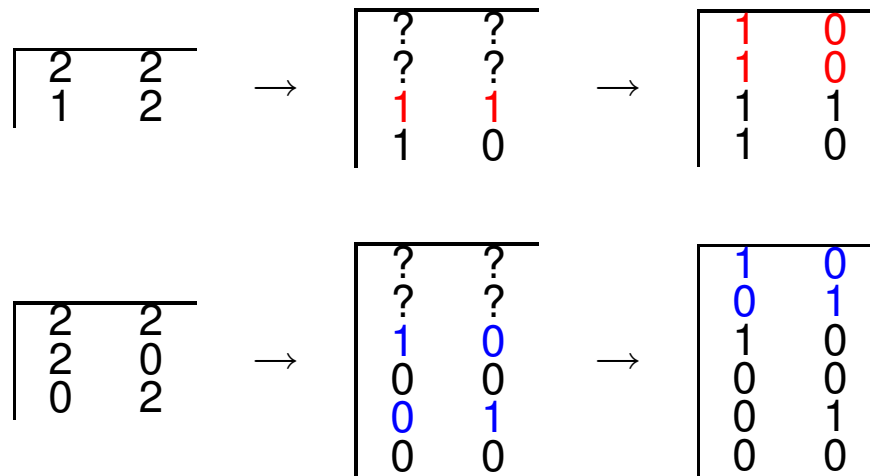
- Let M' be a PPH of M . Cloumns j_{c1}, j_{c2} are *compansions*.
- Row i *equates* j_{c1}, j_{c2} if $M'[i, j_{c1}] = M'[i, j_{c1}]$.
Otherwise, row i *negates* j_{c1}, j_{c2} .
- If a row i *equates* (*negates*) j_{c1}, j_{c2} , then all rows are forced to be *equated* (*negated*).

$$\begin{array}{|c|c|} \hline 2 & 2 \\ \hline 2 & 2 \\ \hline \end{array} \rightarrow \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 0 & 0 \\ ? & ? \\ ? & ? \\ \hline \end{array} ; \begin{array}{|c|c|} \hline 2 & 2 \\ \hline 2 & 2 \\ \hline \end{array} \rightarrow \begin{array}{|c|c|} \hline 1 & 0 \\ \hline 0 & 1 \\ ? & ? \\ ? & ? \\ \hline \end{array}$$

- How fast can we find out all compansion columns?

Forced Expansions (Contd.)

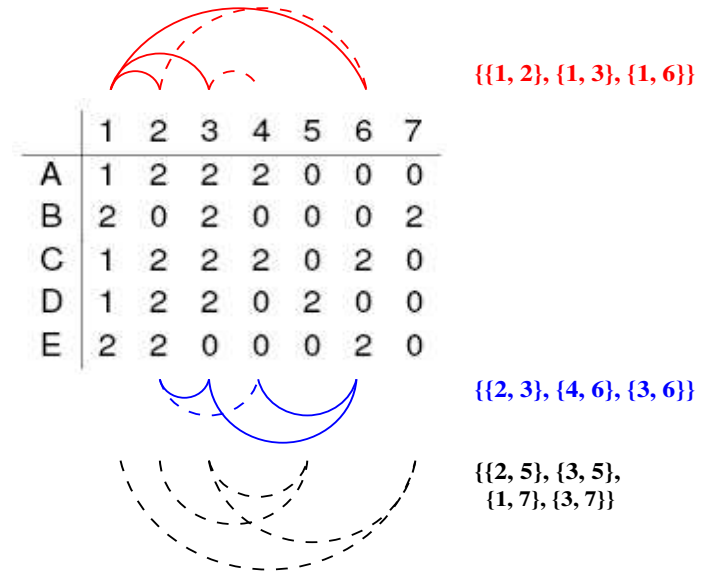
● Observations



● How fast can we find out all forced relationships?

An Example

	1	2	3	4	5	6	7
A	1	2	2	2	0	0	0
B	2	0	2	0	0	0	2
C	1	2	2	2	0	2	0
D	1	2	2	0	2	0	0
E	2	2	0	0	0	2	0



	1	2	3	4	6	5	7
a	1	2	2	2	0	0	0
a'	1	2	2	2	0	0	0
b	2	0	2	0	0	0	2
b'	2	0	2	0	0	0	2
c	1	2	2	2	2	0	0
c'	1	2	2	2	2	0	0
d	1	2	2	0	0	2	0
d'	1	2	2	0	0	2	0
e	2	2	0	0	2	0	0
e'	2	2	0	0	2	0	0

→

	1	2	3	4	6	5	7
a	1	1	0	0	0	0	0
a'	1	0	1	1	0	0	0
b	1	0	1	0	0	0	2
b'	0	0	0	0	0	0	2
c	1	1	0	0	1	0	0
c'	1	0	1	1	0	0	0
d	1	1	0	0	0	2	0
d'	1	0	1	0	0	2	0
e	1	1	0	0	1	0	0
e'	0	0	0	0	0	0	0

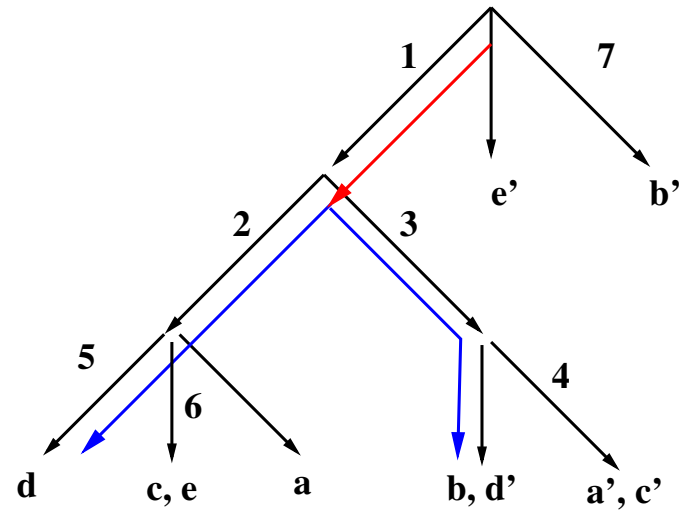
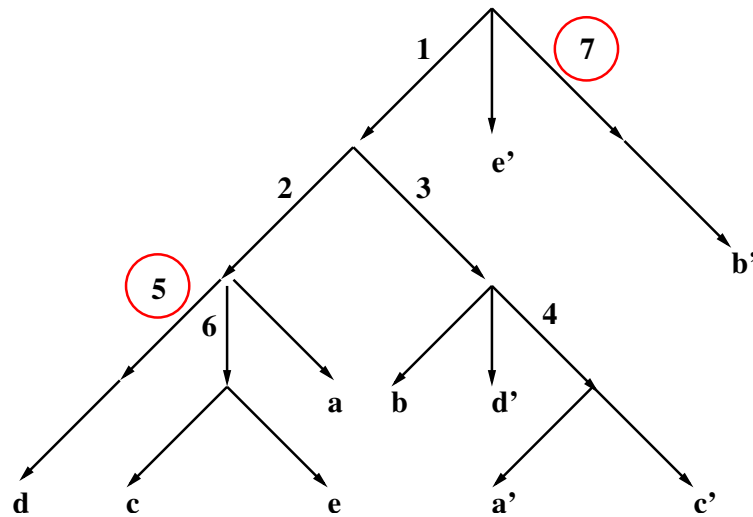
The Solutions

	1	2	3	4	5	6	7
A	1	2	2	2	0	0	0
B	2	0	2	0	0	0	2
C	1	2	2	2	0	2	0
D	1	2	2	0	2	0	0
E	2	2	0	0	0	2	0

→

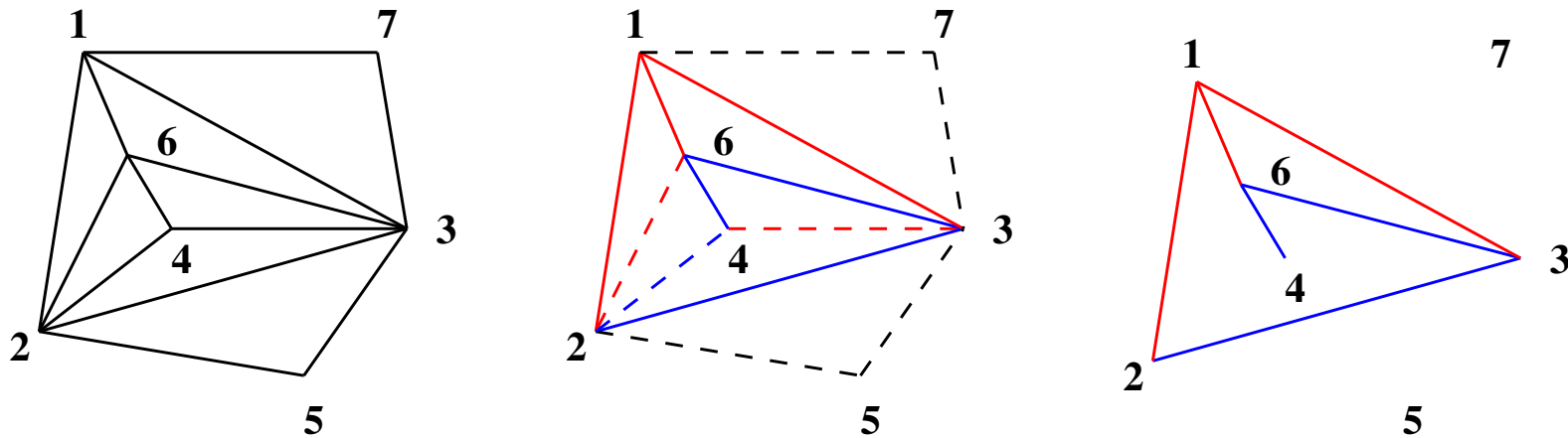
	1	2	3	4	6	5	7	
a	1	1	0	0	0	0	0	1, 2
a'	1	0	1	1	0	0	0	1, 3, 4
b	1	0	1	0	0	0	0	1, 3
b'	0	0	0	0	0	0	1	7
c	1	1	0	0	1	0	0	1, 2, 6
c'	1	0	1	1	0	0	0	1, 3, 4
d	1	1	0	0	0	1	0	1, 2
d'	1	0	1	0	0	0	0	1, 2, 5
e	1	1	0	0	1	0	0	1, 2, 6
e'	0	0	0	0	0	0	0	∅
LC	8	4	4	2	2	1	1	

, ...



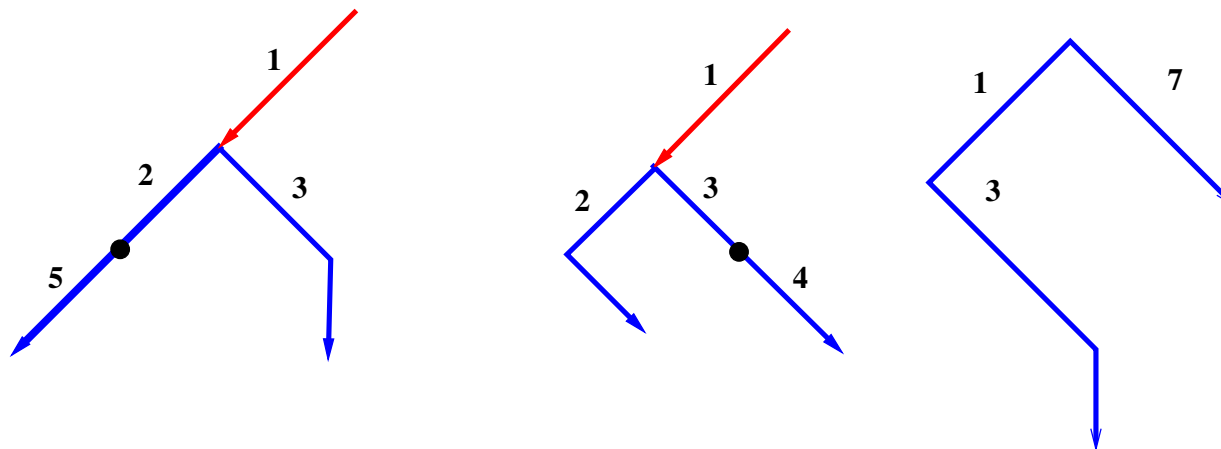
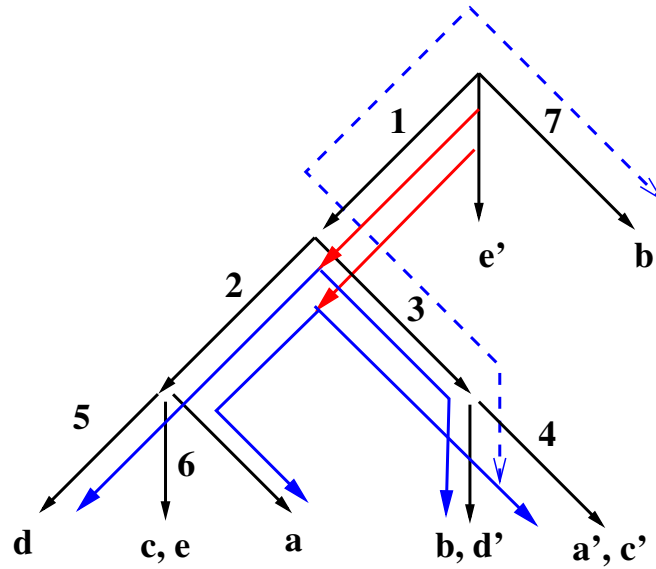
Corollary

● Graph G , G_c and G_f :



- Corollary 1: In a single connected component C of G with k connected components in G_f , there are exactly 2^{k-1} different solutions to the PPH problem in the columns of M represented by C .
- Corollary 2: If G has r connected components and k connected components of G_f , then there are exactly 2^{r-k} solutions to the PPH problem.

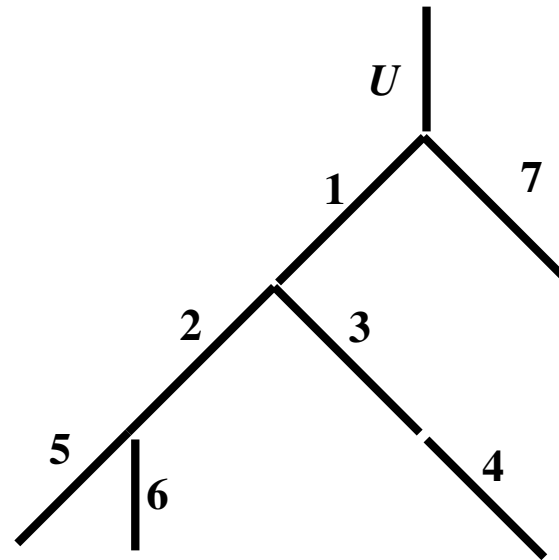
Observations Again



The Graph Relaiization Problem

● Graph Realization Example:

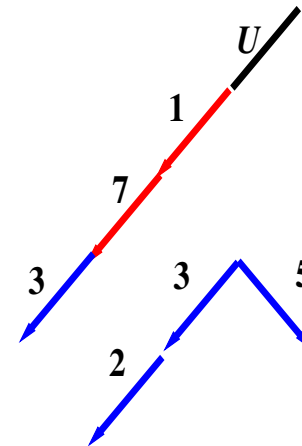
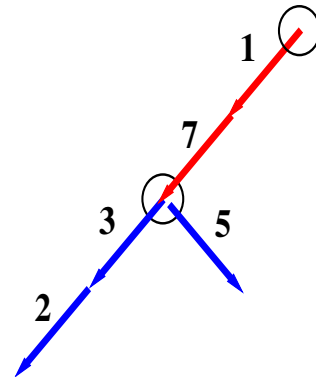
P1: $\{U, 1\}$
P2: $\{U, 1, 2\}$
P3: $\{1, 2\}$
P4: $\{2, 3, 4\}$
P5: $\{1, 3, 7\}$
P6: $\{2, 3, 4, 6\}$
P7: $\{2, 3, 5\}$
P8: $\{1, 2, 6\}$



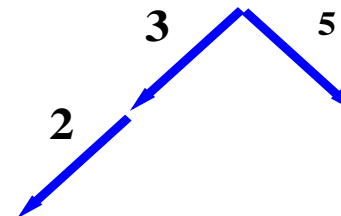
● R. E. Bixby and D. K. Wagner. *An almost linear-time algorithm for graph realization*. 1988

Reducing PPH to Graph Relization

● Encoding a fork



- P1: $\{U, 1\}$
- P2: $\{U, 1, 7\}$
- P3: $\{1, 7\}$
- P4: $\{1, 7, 3\}$
- P5: $\{7, 3\}$



$P : \{2, 3, 5\}$

Redoing the Example

	1	2	3	4	5	6	7
A	1	2	2	2	0	0	0
B	2	0	2	0	0	0	2
C	1	2	2	2	0	2	0
D	1	2	2	0	2	0	0
E	2	2	0	0	0	2	0

→

- P1: $\{U, 1\}$
- P2: $\{U, 1, 2\}$
- P3: $\{1, 2\}$
- P4: $\{2, 3, 4\}$
- P5: $\{1, 3, 7\}$
- P6: $\{2, 3, 4, 6\}$
- P7: $\{2, 3, 5\}$
- P8: $\{1, 2, 6\}$

